

1. **Design a Finite Automata (FA) that accepts all strings over $S=\{0, 1\}$ having three consecutive 1's as a substring. Write a program to simulate this FA.**

```
#include <iostream>
#include <string>
using namespace std;

void state1(string w, int i);
void state2(string w, int i);
void state3(string w, int i);
void state4(string w, int i);

void state1(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='1'){
        state2(w, i+1);
    }
    if(w[i]=='0'){
        state1(w, i+1);
    }
}

void state2(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='1'){
        state3(w, i+1);
    }
    if(w[i]=='0'){
        state1(w, i+1);
    }
}

void state3(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='1'){
        state4(w, i+1);
    }
    if(w[i]=='0'){
        state1(w, i+1);
    }
}
```

```

void state4(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='0' || w[i]=='1'){
        state4(w, i+1);
    }
}

int main(){
    string w;
    cout<<"enter the word: "<<endl;
    cin>>w;
    state1(w, 0);
}

```

OUTPUT:

```

enter the word:
01110
string is accepted

```

```

enter the word:
01100
string is rejected

```

2. Design a Finite Automata (FA) that accepts all strings over $S=\{0, 1\}$ having either exactly two 1's or exactly three 1's, not more nor less. Write a program to simulate this FA.

```
#include <iostream>
#include <string>
using namespace std;

void state1(string w, int i);
void state2(string w, int i);
void state3(string w, int i);
void state4(string w, int i);
void state5(string w, int i);

void state1(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='1'){
        state2(w, i+1);
    }
    if(w[i]=='0'){
        state1(w, i+1);
    }
}

void state2(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='1'){
        state3(w, i+1);
    }
    if(w[i]=='0'){
        state2(w, i+1);
    }
}

void state3(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='1'){
        state4(w, i+1);
    }
    if(w[i]=='0'){
        state3(w, i+1);
    }
}

void state4(string w, int i){
```

```

    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='1'){
        state5(w, i+1);
    }
    if(w[i]=='0'){
        state4(w, i+1);
    }
}

void state5(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='0' || w[i]=='1'){
        state5(w, i+1);
    }
}

int main(){
    string w;
    cout<<"enter the word: "<<endl;
    cin>>w;
    state1(w, 0);
}

```

OUTPUT:

```

enter the word:
01010
string is accepted

```

```

enter the word:
01111
string is rejected

```

3. Design a Finite Automata (FA) that accepts language L1, over S={a, b}, comprising of all strings (of length 4 or more) having first two characters same as the last two. Write a program to simulate this FA.

```
#include <iostream>
#include <string>
using namespace std;

void state1(string w, int i);
void state2(string w, int i);
void state3(string w, int i);
void state4(string w, int i);
void state5(string w, int i);
void state6(string w, int i);
void state7(string w, int i);
void state8(string w, int i);
void state9(string w, int i);
void state10(string w, int i);
void state11(string w, int i);
void state12(string w, int i);
void state13(string w, int i);
void state14(string w, int i);
void state15(string w, int i);

void state1(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state2(w, i+1);
    }
    if(w[i]=='b'){
        state3(w, i+1);
    }
}

void state2(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state4(w, i+1);
    }
    if(w[i]=='b'){
        state5(w, i+1);
    }
}

void state3(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state6(w, i+1);
    }
    if(w[i]=='b'){
        state7(w, i+1);
    }
}

void state4(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state8(w, i+1);
    }
    if(w[i]=='b'){
        state9(w, i+1);
    }
}

void state5(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state10(w, i+1);
    }
    if(w[i]=='b'){
        state11(w, i+1);
    }
}

void state6(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state12(w, i+1);
    }
    if(w[i]=='b'){
        state13(w, i+1);
    }
}

void state7(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state14(w, i+1);
    }
    if(w[i]=='b'){
        state15(w, i+1);
    }
}

void state8(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state1(w, i+1);
    }
    if(w[i]=='b'){
        state2(w, i+1);
    }
}

void state9(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state3(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state10(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state5(w, i+1);
    }
    if(w[i]=='b'){
        state6(w, i+1);
    }
}

void state11(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state7(w, i+1);
    }
    if(w[i]=='b'){
        state8(w, i+1);
    }
}

void state12(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state9(w, i+1);
    }
    if(w[i]=='b'){
        state10(w, i+1);
    }
}

void state13(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state11(w, i+1);
    }
    if(w[i]=='b'){
        state12(w, i+1);
    }
}

void state14(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state13(w, i+1);
    }
    if(w[i]=='b'){
        state14(w, i+1);
    }
}

void state15(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state15(w, i+1);
    }
    if(w[i]=='b'){
        state15(w, i+1);
    }
}
```

```

    }
    if(w[i]=='a'){
        state6(w, i+1);
    }
    if(w[i]=='b'){
        state7(w, i+1);
    }
}

void state4(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state8(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state5(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state10(w, i+1);
    }
    if(w[i]=='b'){
        state5(w, i+1);
    }
}

void state6(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state6(w, i+1);
    }
    if(w[i]=='b'){
        state12(w, i+1);
    }
}

void state7(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){

```

```

        state7(w, i+1);
    }
    if(w[i]=='b'){
        state14(w, i+1);
    }
}

void state8(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state9(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state9(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state8(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state10(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state10(w, i+1);
    }
    if(w[i]=='b'){
        state11(w, i+1);
    }
}

void state11(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state10(w, i+1);
    }
}

```

```

        if(w[i]=='b'){
            state5(w, i+1);
        }
    }

void state12(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state13(w, i+1);
    }
    if(w[i]=='b'){
        state12(w, i+1);
    }
}

void state13(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state6(w, i+1);
    }
    if(w[i]=='b'){
        state12(w, i+1);
    }
}

void state14(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state7(w, i+1);
    }
    if(w[i]=='b'){
        state15(w, i+1);
    }
}

void state15(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state7(w, i+1);
    }
    if(w[i]=='b'){
        state14(w, i+1);
    }
}

```

```
    }  
}  
  
int main(){  
    string w;  
    cout<<"enter the word: "<<endl;  
    cin>>w;  
    state1(w, 0);  
}
```

OUTPUT:

```
enter the word:  
aabaa  
string is accepted
```

```
enter the word:  
aabb  
string is rejected
```

4. Design a Finite Automata (FA) that accepts language L_2 , over $S = \{a, b\}$ where $L_2 = a(a+b)^*b$. Write a program to simulate this FA.

```
#include <iostream>
#include <string>
using namespace std;

void state1(string w, int i);
void state2(string w, int i);
void state3(string w, int i);
void state4(string w, int i);

void state1(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state2(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state2(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state2(w, i+1);
    }
    if(w[i]=='b'){
        state3(w, i+1);
    }
}

void state3(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state2(w, i+1);
    }
    if(w[i]=='b'){
        state3(w, i+1);
    }
}

void state4(string w, int i){
    if(i==w.length()){
```

```
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a' || w[i]=='b'){
        state4(w, i+1);
    }
}

int main(){
    string w;
    cout<<"enter the word: "<<endl;
    cin>>w;
    state1(w, 0);
}
```

OUTPUT:

```
enter the word:
aabab
string is accepted
```

```
enter the word:
baba
string is rejected
```

5. **Design a Finite Automata (FA) that accepts language EVEN-EVEN over $S=\{a, b\}$. Write a program to simulate this FA.**

```
#include <iostream>
#include <string>
using namespace std;

void state1(string w, int i);
void state2(string w, int i);
void state3(string w, int i);
void state4(string w, int i);

void state1(string w, int i){
    if(i==w.length()){
        cout<<"string is accepted"<<endl;
        return;
    }
    if(w[i]=='a'){
        state2(w, i+1);
    }
    if(w[i]=='b'){
        state3(w, i+1);
    }
}

void state2(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state1(w, i+1);
    }
    if(w[i]=='b'){
        state4(w, i+1);
    }
}

void state3(string w, int i){
    if(i==w.length()){
        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state4(w, i+1);
    }
    if(w[i]=='b'){
        state1(w, i+1);
    }
}

void state4(string w, int i){
    if(i==w.length()){
```

```

        cout<<"string is rejected"<<endl;
        return;
    }
    if(w[i]=='a'){
        state3(w, i+1);
    }
    if(w[i]=='b'){
        state2(w, i+1);
    }
}

int main(){
    string w;
    cout<<"enter the word: "<<endl;
    cin>>w;
    state1(w, 0);
}

```

OUTPUT:

```

enter the word:
aabb
string is accepted

```

```

enter the word:
aaab
string is rejected

```

6. Write a program to simulate an FA that accepts

a. Union of the languages L1 and L2

b. Intersection of the languages L1 and L2

c. Language L1 L2 (concatenation)

```
#include <iostream>
#include <string>
using namespace std;

bool FA1(string w){
    if(w.empty()) return false;
    return (w.back()=='a');
}

bool FA2(string w){
    if(w.empty()) return false;
    return (w.front()=='b');
}

bool UnionFA(string w){
    return (FA1(w) || FA2(w));
}

bool IntersectionFA(string w){
    return (FA1(w) && FA2(w));
}

bool ConcatenationFA(string w){
    int n=w.length();
    for(int k=0; k<=n; k++){
        string x=w.substr(0,k);
        string y=w.substr(k);
        if(FA1(x) && FA2(y))
            return true;
    }
    return false;
}

int main(){
    string w;
    cout<<"Enter a string (over {a,b}): ";
    cin>>w;
    cout<<"\nResult of simulation:\n";
    cout<<"L1 (ends with a): "<<(FA1(w)?"Accepted":"Rejected")<<endl;
    cout<<"L2 (starts with b): "<<(FA2(w)?"Accepted":"Rejected")<<endl;
    cout<<"\n(a) Union (L1 U L2): "<<(UnionFA(w)?"Accepted":"Rejected")<<endl;
    cout<<"(b) Intersection (L1 n L2): "<<(IntersectionFA(w)?"Accepted":"Rejected")<<endl;
    cout<<"(c) Concatenation (L1L2): "<<(ConcatenationFA(w)?"Accepted":"Rejected")<<endl;
    return 0;
}
```

OUTPUT:

Enter a string (over {a,b}): ba

Result of simulation:

L1 (ends with a): Accepted

L2 (starts with b): Accepted

(a) Union ($L1 \cup L2$): Accepted

(b) Intersection ($L1 \cap L2$): Accepted

(c) Concatenation ($L1L2$): Rejected